

The paper

Proving what a second-hand computer is: an evidence-first architecture for IT asset disposition

A complete technical paper on the whole project — every layer, what was achieved, what was rejected, and what is still unproven. 78 days, 535 commits, ~101,000 lines.

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Read this first

File	What it is
PAPER.md	The whole paper, assembled. Start here.
OUTLINE.md	The structure and the rules this draft was written to
assemble.py	Regenerates PAPER.md and counts prose and tables

sections/ holds one file per section — edit those, never PAPER.md. Each section file ends with **drafting notes**: open questions, things to verify, and a record of what was corrected and why. Those are excluded from PAPER.md and from the word counts.

The nineteen sections

Part	§	Section	Words
	—	Abstract	324
I. The problem	1	Introduction	843
	2	Requirements and constraints	670
	3	System overview	683
II. The layers	4	The inventory model	814
	5	Offline-first capture	904
	6	The audit station	1,113
	7	Deep inspection	1,062
	8	Erasure and certification	1,084
	9	Functional testing	671
	10	The back office	865
III. What it taught	11	The knowability framework	1,011
	12	Evidence integrity: the false absence	1,066
	13	The discipline of deletion	833
	14	Engineering practice	962
IV. Closing	15	Results	403
	16	Limitations and threats	722
	17	Lessons	971
	18	Related work	648
	19	Conclusion	529

16,178 words of prose, 29 tables and figures. Regenerate the figures with `python paper/assemble.py` — do not hand-edit them into this file.

The three findings

1. **Knowability tiers** (§11) — claims about external state are measured, recovered from an artefact, perishable, or withheld by a third party, and the tier decides whether code, a process change, or nothing will produce an answer. The most valuable question in this domain sits in the fourth tier.
2. **The false absence** (§12) — a failed observation reported as a confirmed negative. 32 call sites, all 32 failing towards the reassuring answer, 18 on code paths whose existing tests caught none of them.
3. **Naming a defect class is not a control** (§12.4) — the same author wrote eight fresh instances within three weeks of naming it, documenting it in the code, and adding tests worth a third of the diff.

The evidence

evidence/instances.csv	The defect dataset: 32 rows, 19 fields, one per call site
evidence/inclusion-criteria.md	What counts as an instance, fixed before the analysis
../docs/design/	40 dated design records. The paper is written from these

Every figure in the paper is recomputable from the repository or the dataset.

archive/ — the earlier draft

The first attempt made the false-absence defect class the entire subject. That was the most publishable single claim, not the work. It is kept intact because it holds the full treatment of §12 — the method, the inclusion criteria, the latency distribution, the per-dataset threats — at about 3,700 words, of which §12 is a 1,300-word compression.

It should be offered as a **companion artefact**, not folded back in.

Venue

Not decided, and the earlier arithmetic no longer applies: at ~16,000 words this is a systems paper, so IEEE Software's 4,200-word limit is simply out of scope and is dropped.

Option	Fit
arXiv cs.SE and/or Zenodo	No limit, citable immediately. Zenodo needs no endorsement; arXiv now requires one for new submitters. <i>Recommended first.</i>
ICSE SEIP / ESEM industry track	Refereed, deadline-driven, roughly 6,000–8,000 words — would need cutting, most likely Part II
A short version, later	§11.4 plus §12.4 is a ~2,400-word article. Write it <i>from</i> this, never by hacking this down

Before any submission

- **Verify every citation in §18 against the source.** They are positioned from knowledge of those literatures, not from the papers in hand. This is the one hard blocker. §18's notes list exactly what to check.
- Complete the search for a prior naming of the false-absence class. §18.5 states the framing conditionally for this reason.
- Work through the per-section drafting notes; several name a specific cross-check ("check §17's thesis matches §1.1 word for word").
- Verify §15.3's evidence table against each design record's "How it was proved" heading.

- Decide whether the business is named or anonymised.
- If a second rater can be found, re-rate ten random instances from `evidence/instances.csv`. That single addition answers §16.2, the most serious threat in the paper.
- Re-run `python paper/assemble.py` and re-verify every figure.